Secure long distance QKD

and provably secure steganography

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Part 1: long distance QKD

nature photonics

LETTERS

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Provably secure and practical quantum key distribution over 307 km of optical fibre

Boris Korzh[1]*, Charles Ci Wen Lim[1]*, Raphael Houlmann[1], Nicolas Gisin[1], Ming Jun Li[2], Daniel Nolan[2], Bruno Sanguinetti[1], Rob Thew[1] and Hugo Zbinden[1]

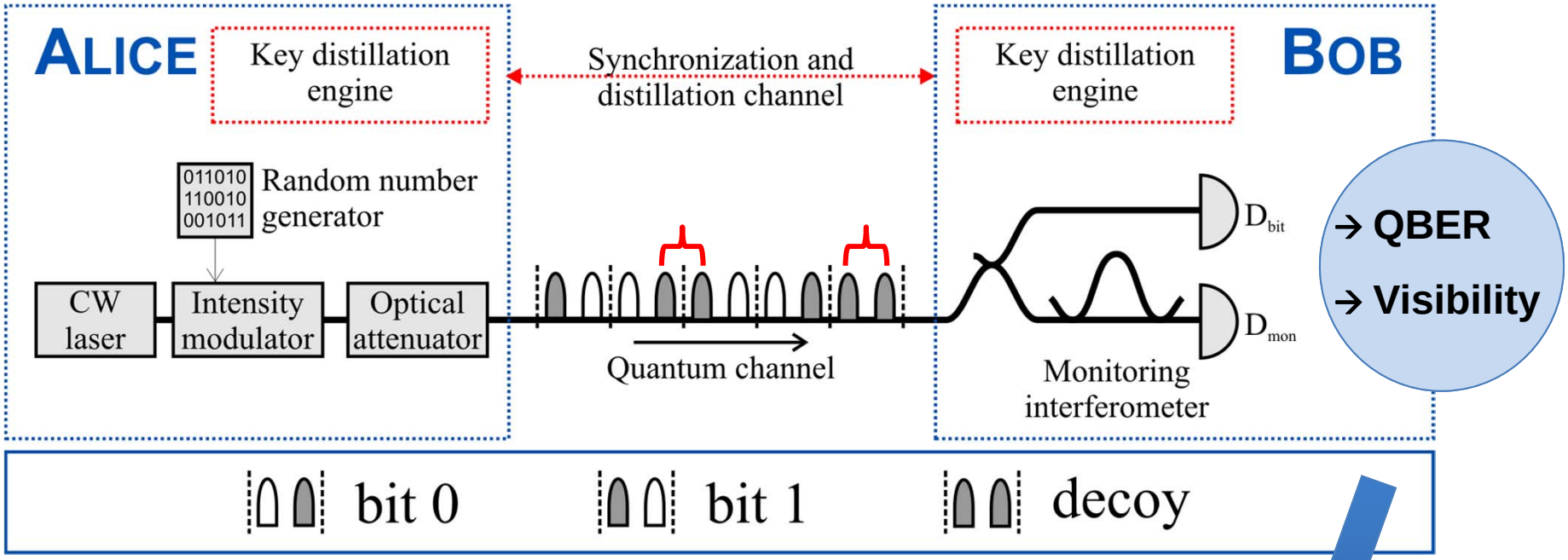
Proposed in 1984, quantum key distribution (QKD) allows two users to exchange provably secure keys via a potentially insecure quantum channel[1]. Since then, QKD has attracted much attention and significant progress has been made both in

sends an additional test state, $|\alpha_t\rangle := |\alpha\rangle|\alpha\rangle$, to check for phase coherence between any two successive laser pulses. Therefore, phase coherence can be checked in any of these sequences, $|\alpha_0\rangle|\alpha_1\rangle$, $|\alpha_0\rangle|\alpha_t\rangle$, $|\alpha_t\rangle|\alpha_1\rangle$, $|\alpha_t\rangle$, $|\alpha_t\rangle|\alpha_t\rangle$, by using an imbalanced

Nature Photonics 9, 163–168 (2015)

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Ingredient 1: efficient and simple QKD scheme



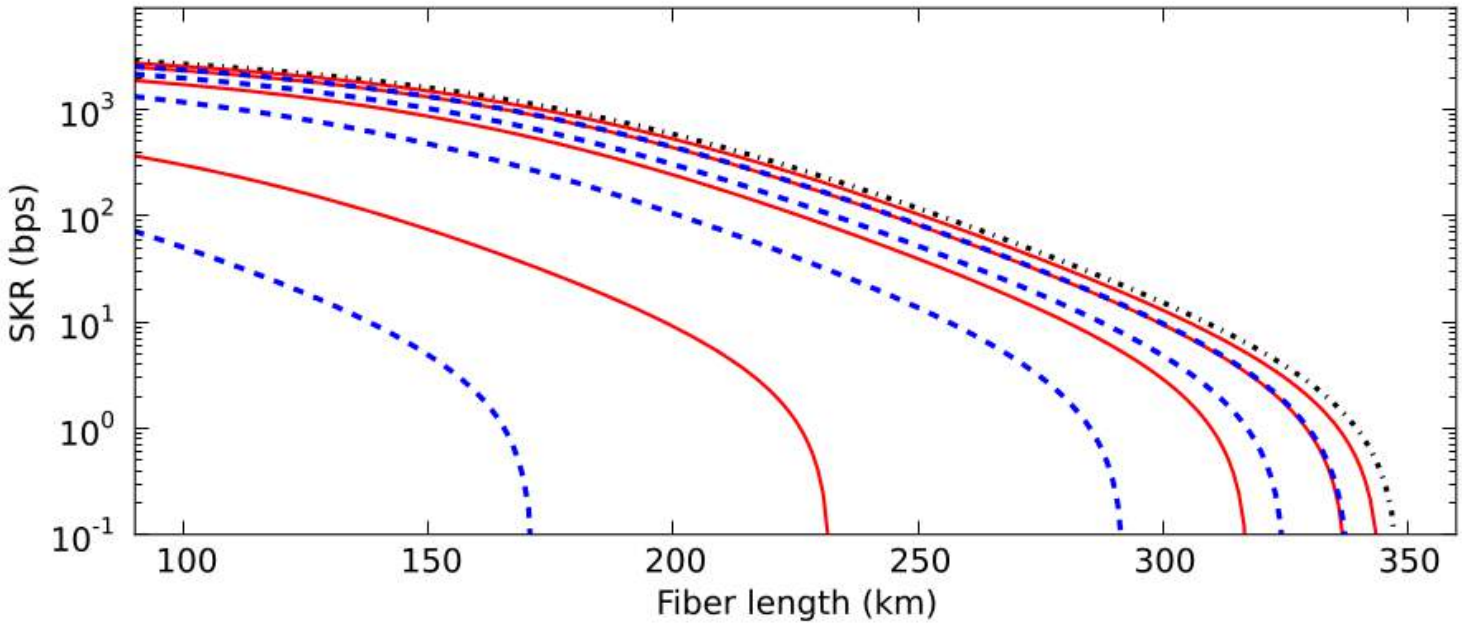
Coherent One Way (COW) Characteristics

- 1.25 GHz clock (625 MHz bit generation rate)
- No active elements at Bob, robust bit measurement basis
- Robust against PNS attacks
- Security proof for collective attacks

Reveals action of eavesdropper
Input for key distillation

Ingredient 2: tight finite key analysis

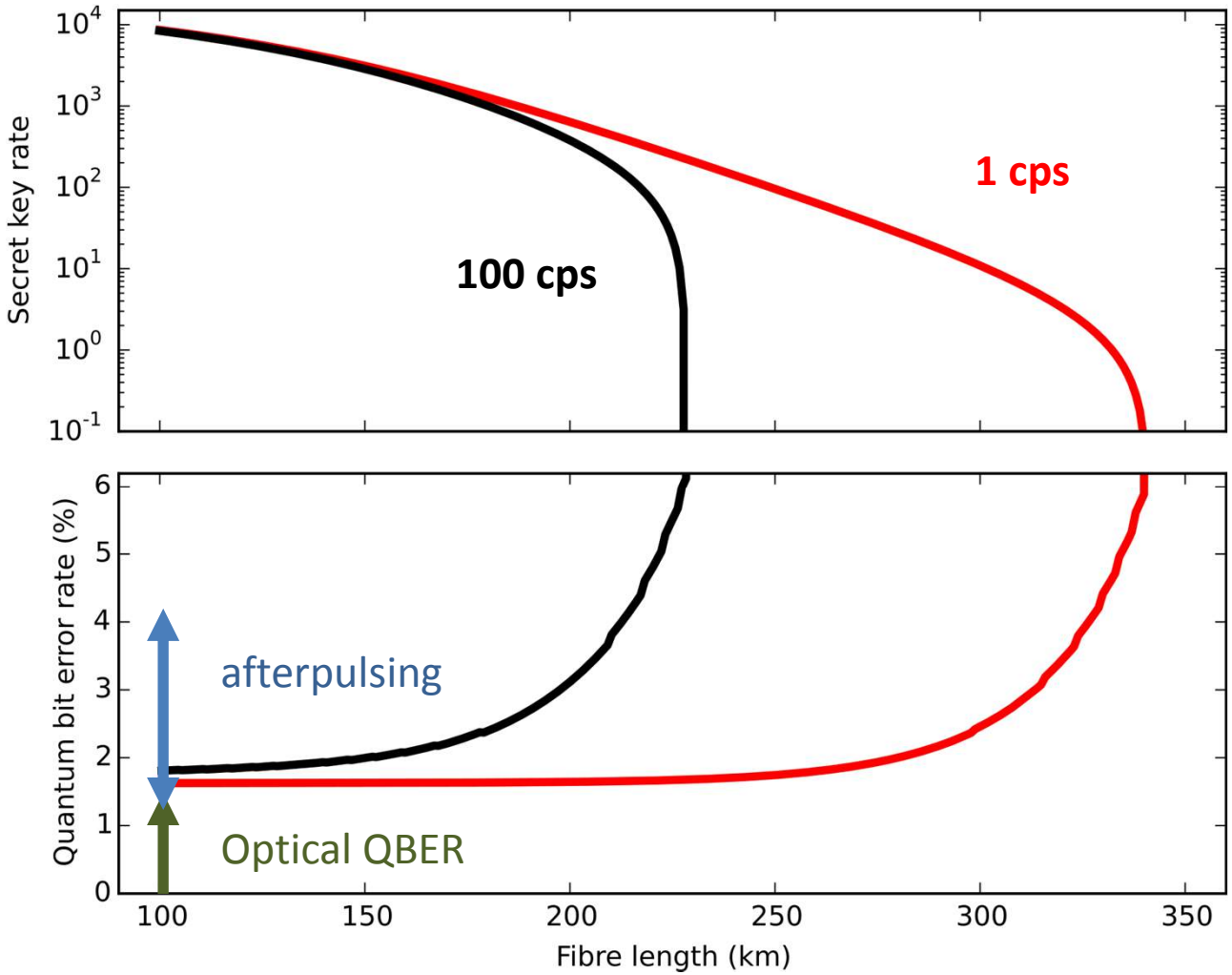
Allows around an order of magnitude reduction
of post-processing block size



Comparison of secret key rate using different postprocessing blocksizes (10^4, 10^5, 10^6, 10^7 left to right)
Solid red: New tail inequality
Dashed blue: Previous tail inequality

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Ingredient 3: low noise single photon detectors



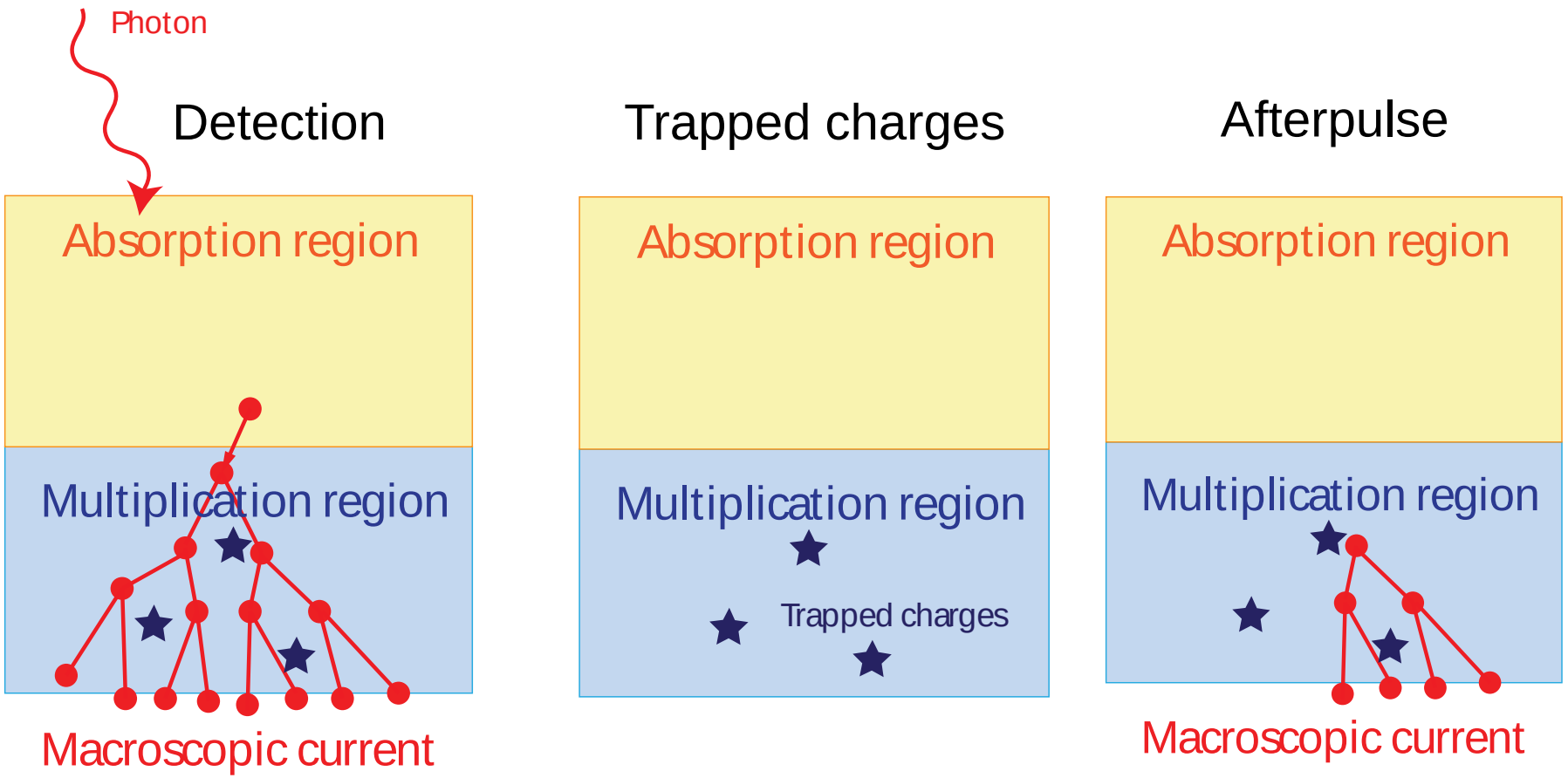
System requirements:

- Low **dark count rate** of SPD
- Compact (no SNSPD)

APDs: afterpulsing!

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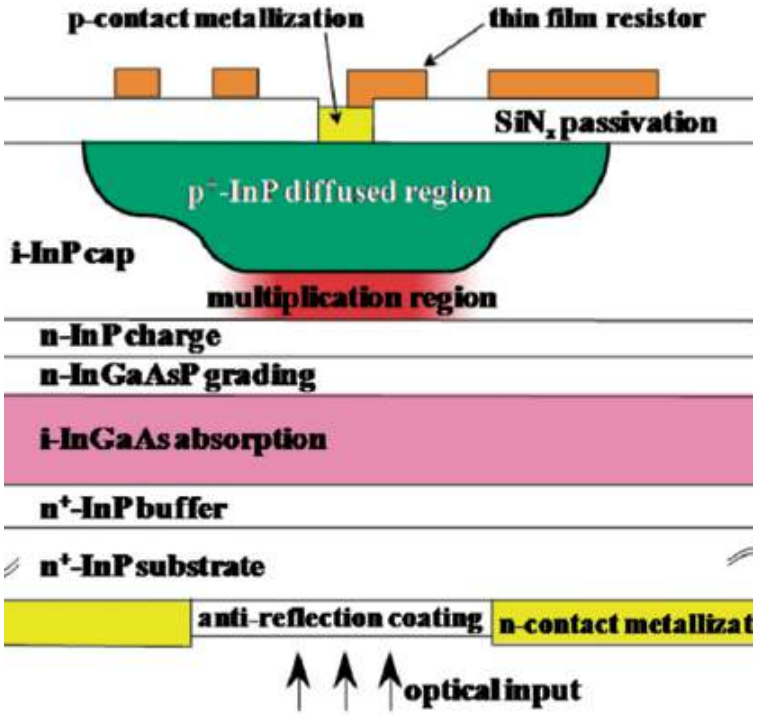
Afterpulsing in APDs

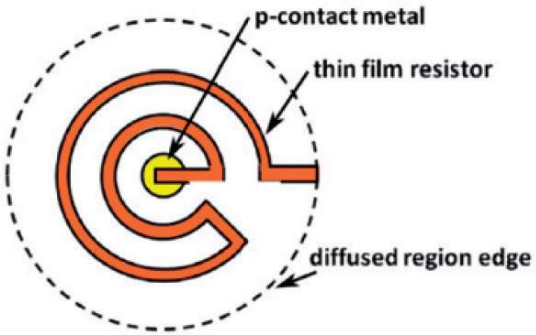


More current flow = More trapped charges = More afterpulsing

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Free running Negative Feedback Aavalanche Photo Diode



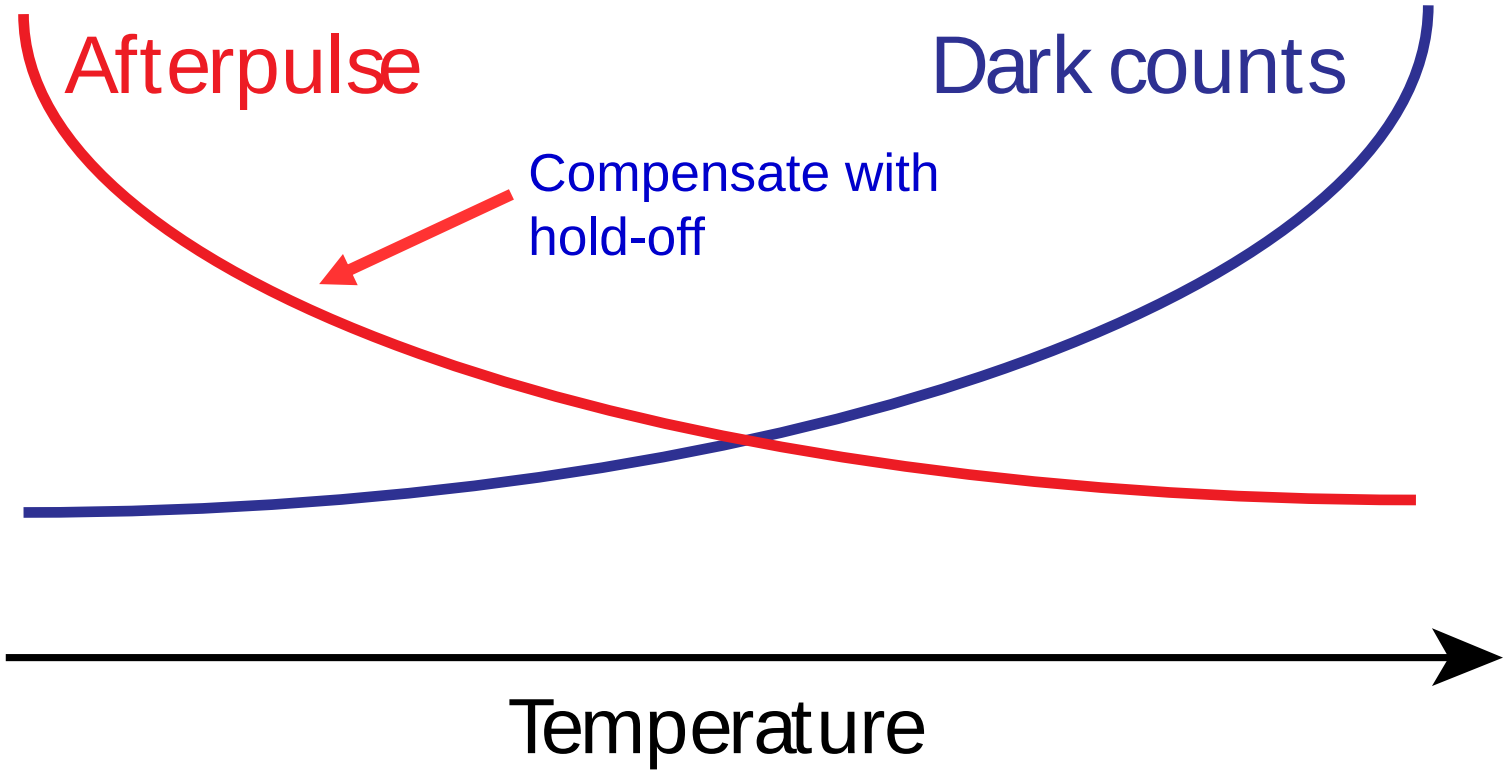


- Rapid passive quenching
  + hold-off time
  -> low afterpulsing
- 1 darkcount/s
  (10% eff, 160 K)

- M. Itzler et al., Proc. SPIE 2009, 7222, 72221K-1
- B. Korzh et al., Appl. Phys. Lett. 104, 081108 (2014)

Tradeoff: Temperature and noise

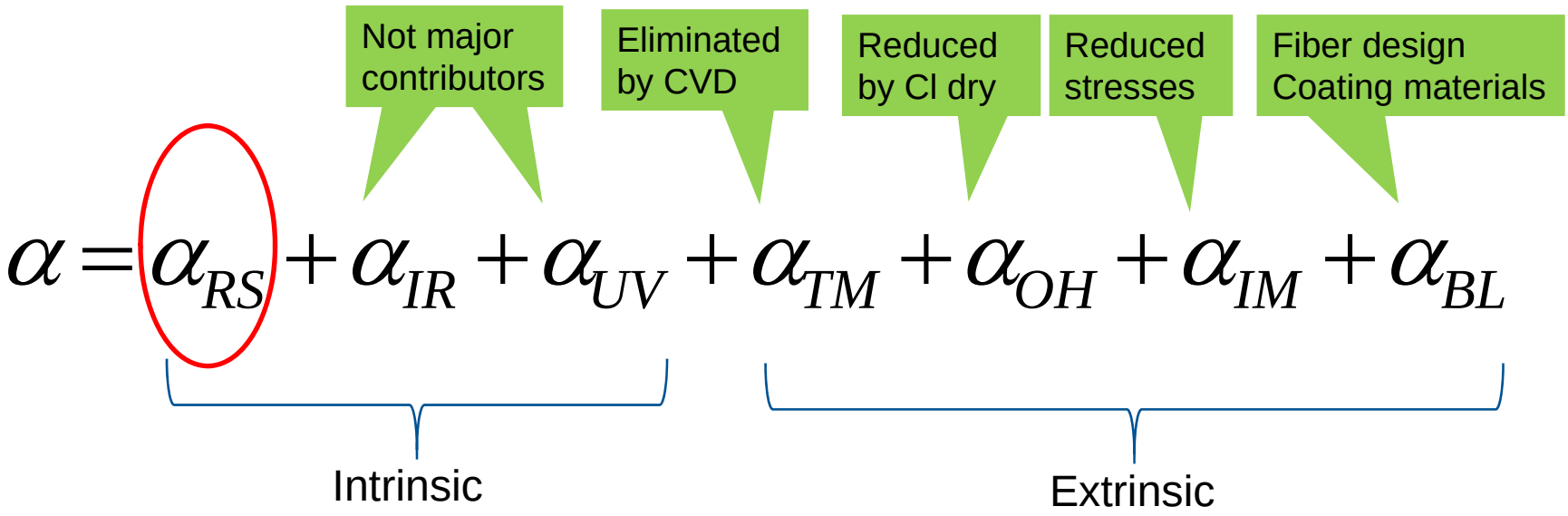
Afterpulse mitigation with longer hold-off time



Ingredient 4: Low Loss Optical Fibres

Total attenuation of an optical fiber:

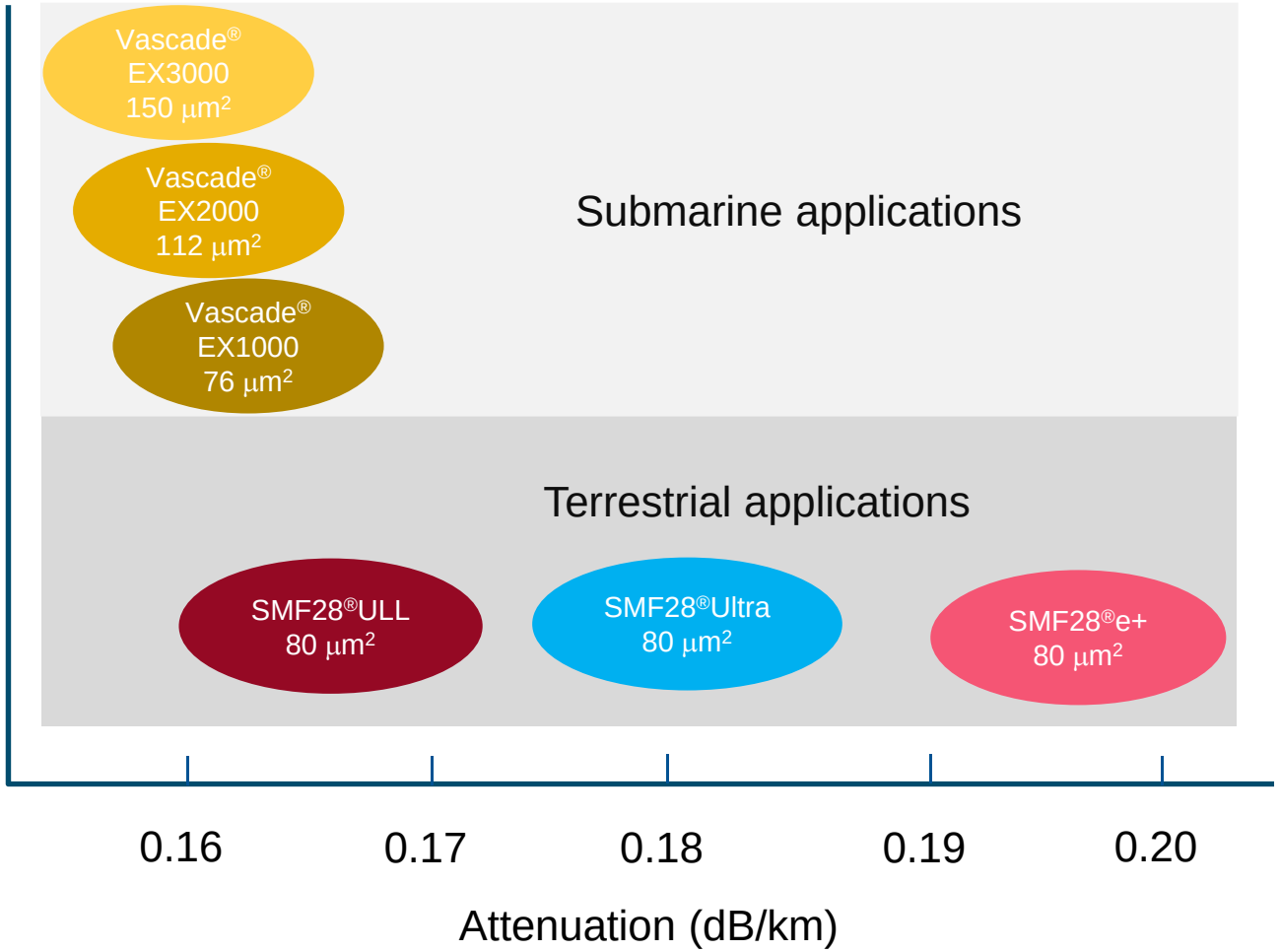
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Rayleigh scattering is dominant: density and dopant fluctuations minimized by choosing optimum (small) dopant concentration.

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Ultra low loss fibers

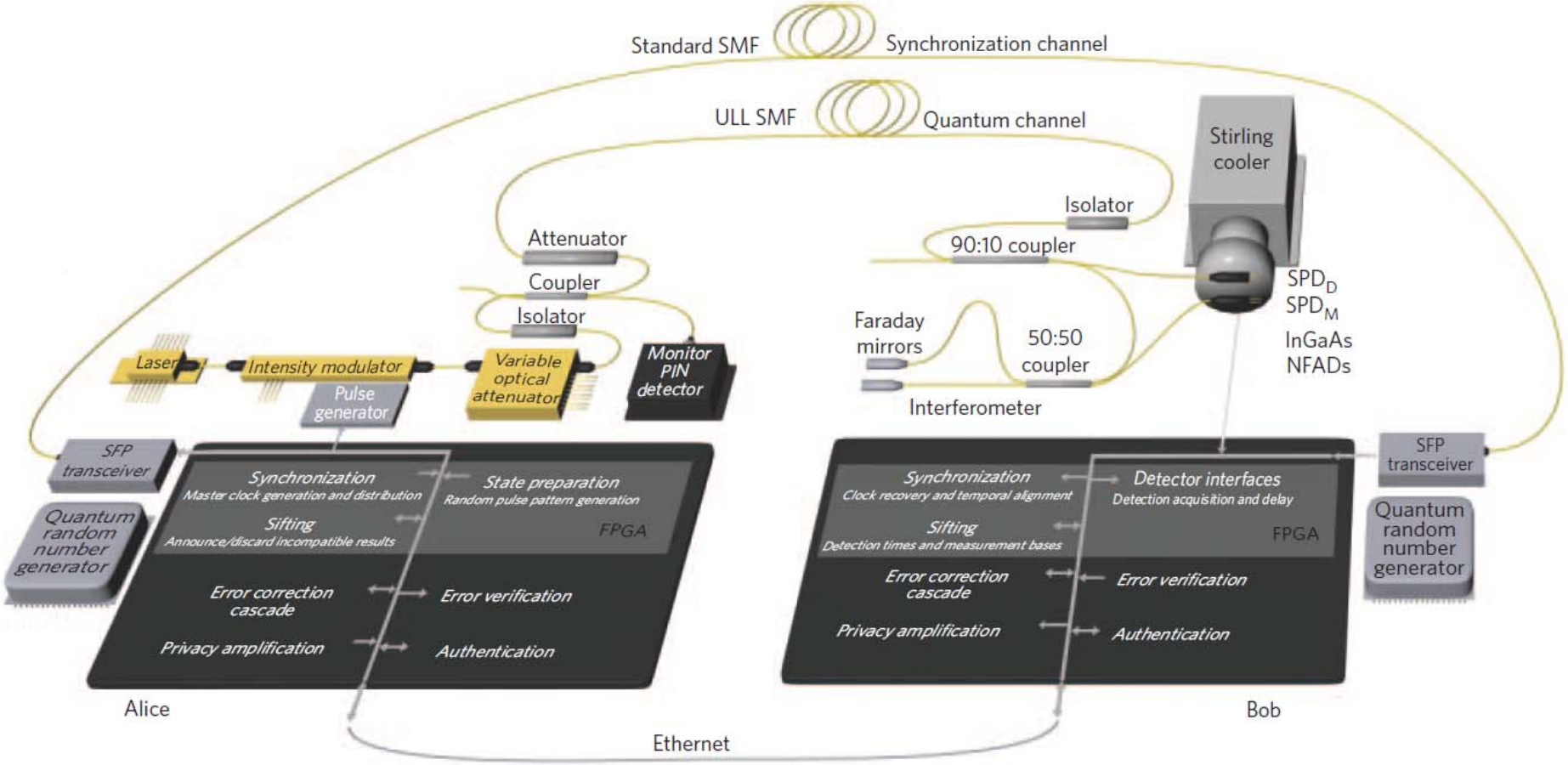


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….putting all together:



FPGA is essential!

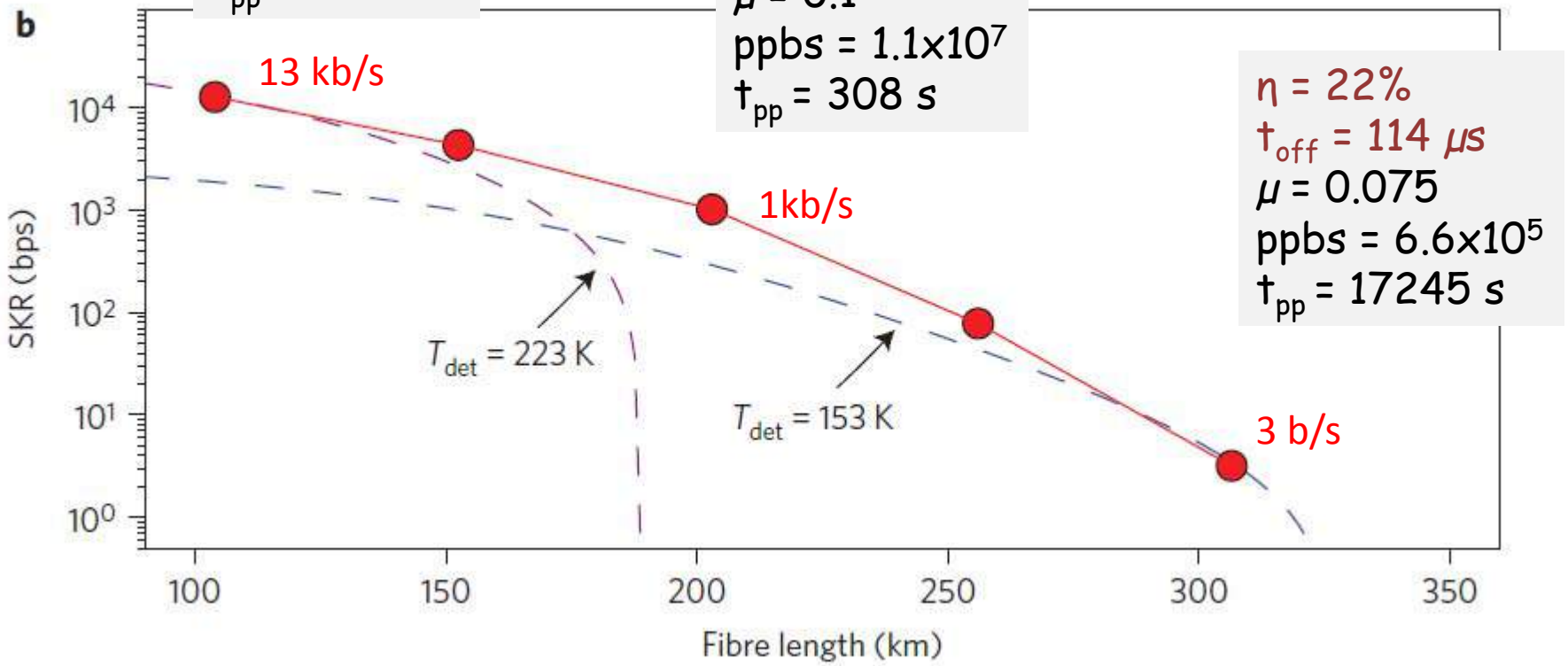
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Results: Secret (finite key) rates vs distance

η = 22%
t_{off} = 9 μs
μ = 0.06
ppbs = 2×10^7
t_{pp} = 537 s

η = 27%
t_{off} = 42 μs
μ = 0.1
ppbs = 1.1×10^7
t_{pp} = 308 s

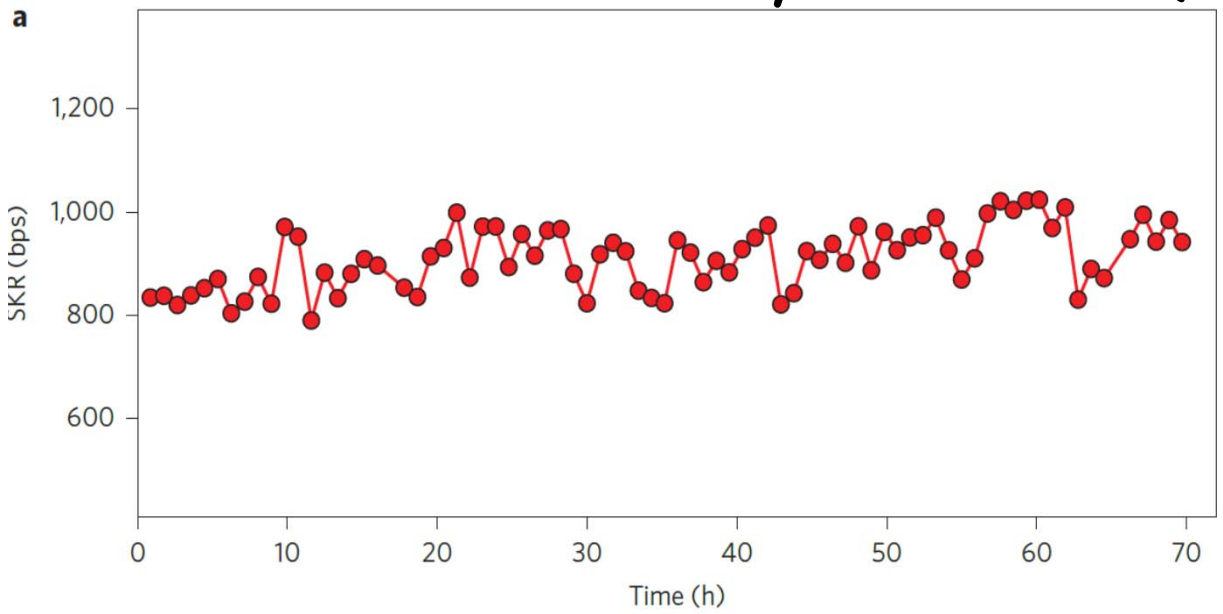
η = 22%
t_{off} = 114 μs
μ = 0.075
ppbs = 6.6×10^5
t_{pp} = 17245 s

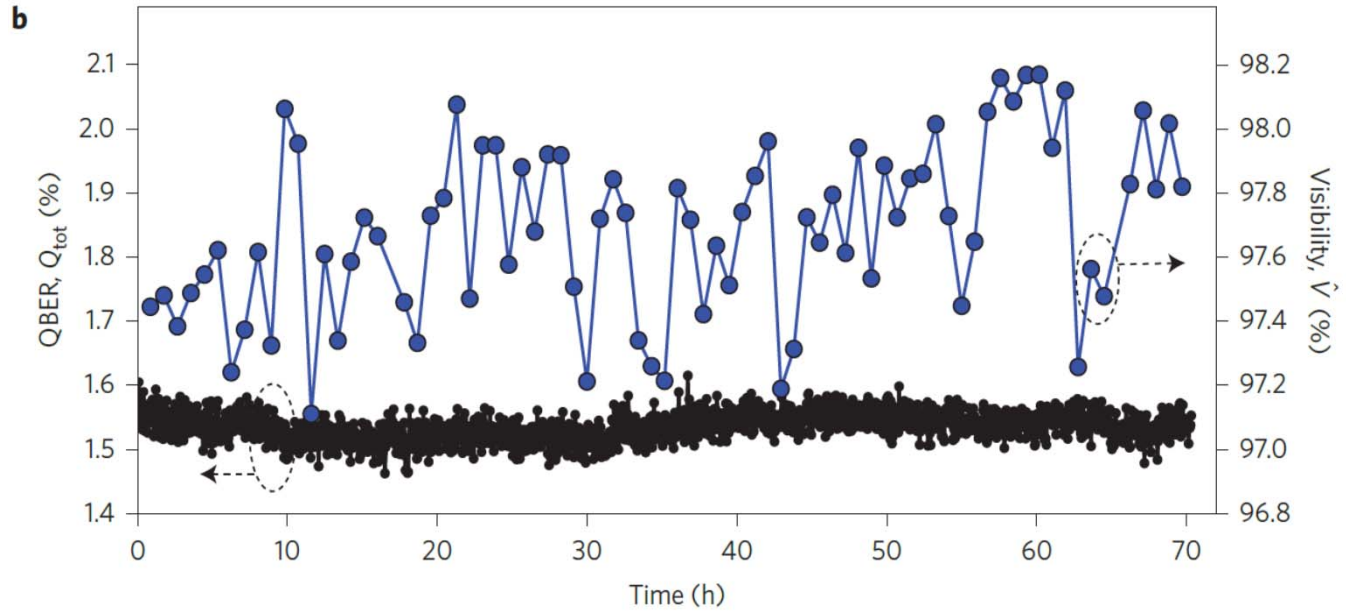


$\varepsilon_{QKD} = 4 \times 10^{-9}$

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Stability over 70h (200km)





Automatic tracking:

QBER

Temporal alignment:
Quantum signal clock recovery with 10 ps resolution

Extinction ratio:
Modulator bias voltage

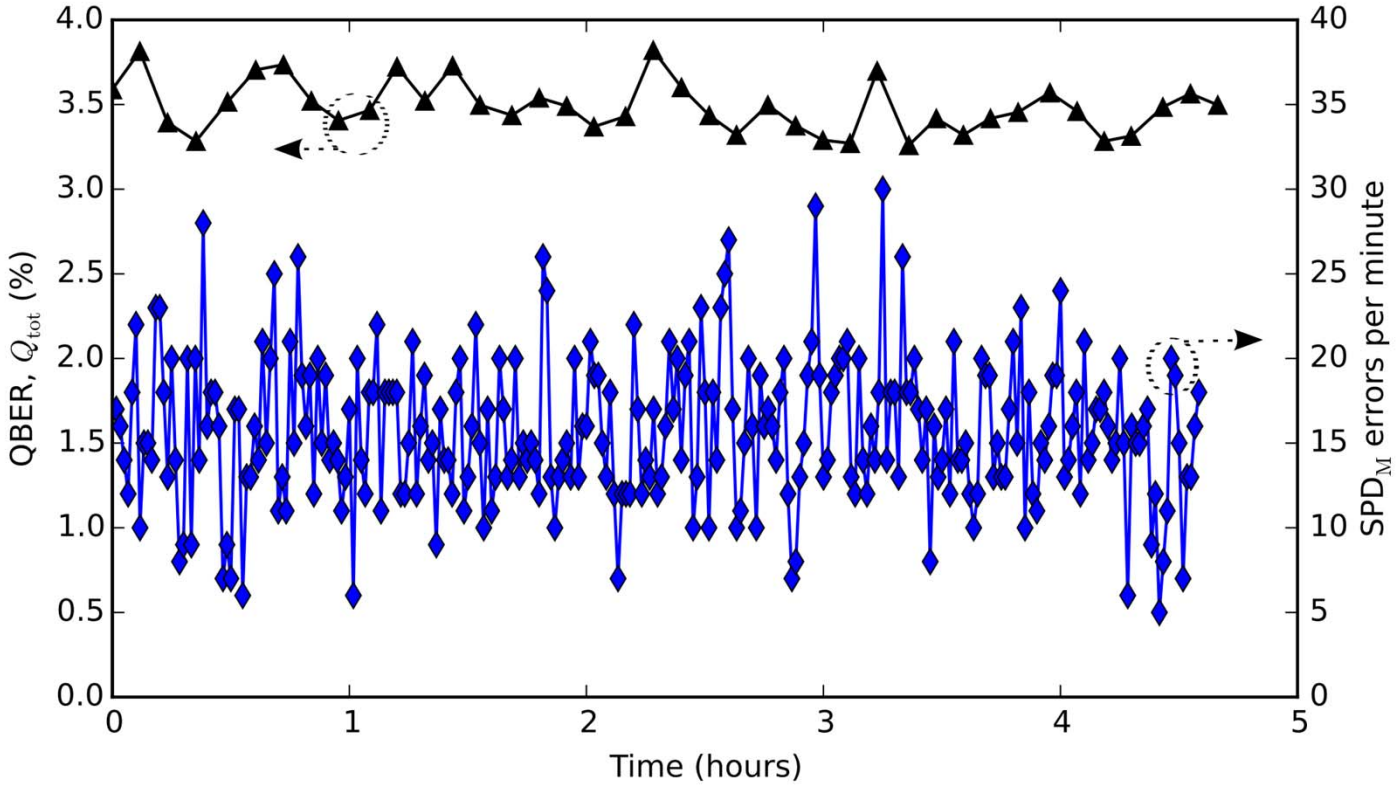
Visibility

Adjust Laser current (wavelength)

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307 km (Collection of 1 data block)

6.6 x 10^5 bits



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Summary: Notable QKD demonstrations

	Quantum channel		Detector			Security			
	Distance (km)	Attenuation (dB)	Type	Temperature (K)	Protocol	Attacks	Finite-key size	ϵ_{qkd}	r_{sec} (bps)
This work	307	51.9	InGaAs‡	153	COW	Collective	6.6×10^5	4×10^{-9}	3.18
Wang 2012[24]	260	52.9	SNSPD	1.7	DPS	Individual	-	-	1.85
Stucki 2009[25]	250	42.9	SNSPD	2.5	COW	Collective	-	-	15
Takesue 2007[26]	200	42.1	SNSPD	3	DPS	Individual	-	-	12.1
Liu 2010[27]	200	-	SNSPD	2.4	BB84	Collective	-	-	15
Rosenberg 2009[28]	135	27.8	SNSPD	3	BB84	Collective	-	-	0.2
Namekata 2011[29]	160	33.6	InGaAs§	193	DPS	Individual	-	-	490
Yuan 2009[30]	100	20	InGaAs§	243	BB84	Collective	-	-	1.01×10^4
Shimizu 2014[23]	90*	30	SNSPD	2.5	DPS	Individual	-	-	1,100
Lucamarini 2013[19]	80†	16	InGaAs§	243	BB84	Collective	10^9	10^{-10}	1.20×10^5
Walenta 2014[18]	25†	5.3	InGaAs§	293	COW	Collective	10^6	4×10^{-9}	2.25×10^4

First long distance experiment with APDs

First long distance experiment with finite key analysis and quantifiable security statement

- B. Korzh, C. W. Lim et al., "*Provably Secure and Practical Quantum Key Distribution over 307 km of Optical Fibre*," Nature Photonics 9, 163-168 (2015)

PART 2:
Quantum Secure Steganography

Perfectly secure steganography: hiding information in the quantum noise of a photograph

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Group of Applied Physics, University of Geneva, Switzerland

Giulia Traverso
Fachbereich Informatik, Technische Universität Darmstadt, Germany

arXiv 1509.07106

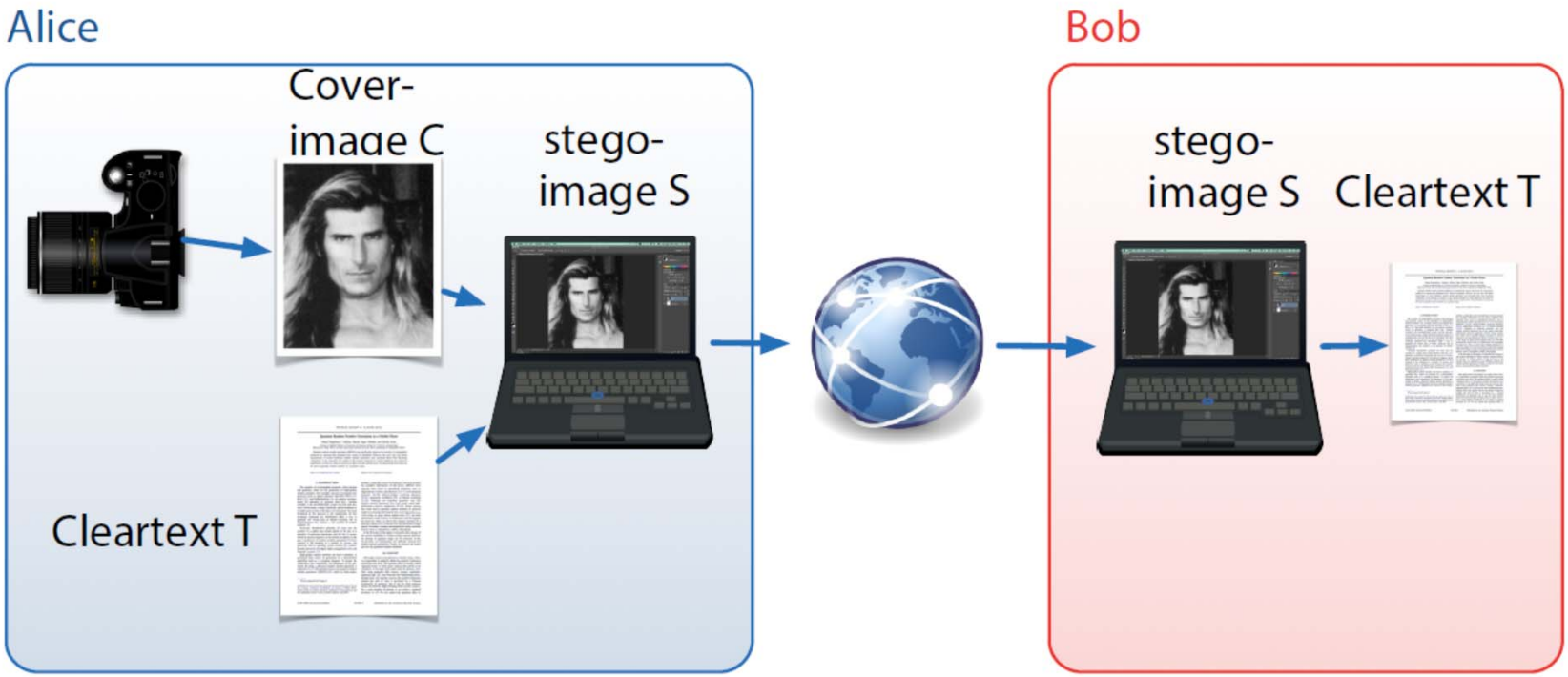
Disclaimer: We are physicists….

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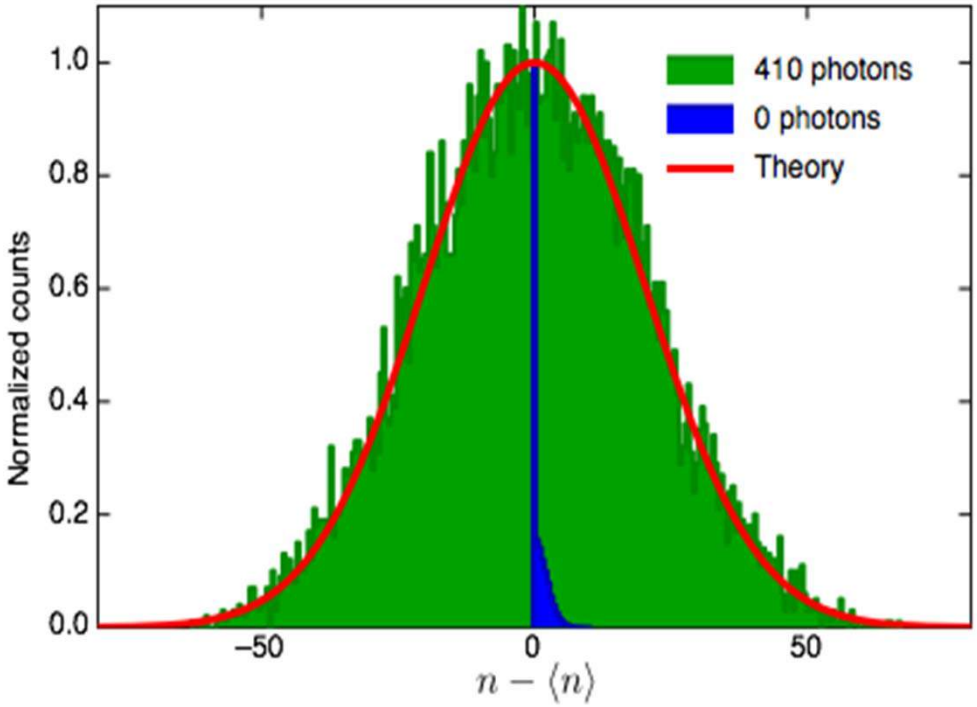
PART 2: Quantum Secure Steganography

- from Greek *steganos*, or "covered," and *graphie*, or "writing"): hiding of a secret message within an ordinary message
- Cryptography guarantees secrecy, but not privacy.
- Steganography important in countries with untrustworthy, totalitarian regimes
- Universal Declaration of Human Rights: Art. 19

Hiding secret information in a picture

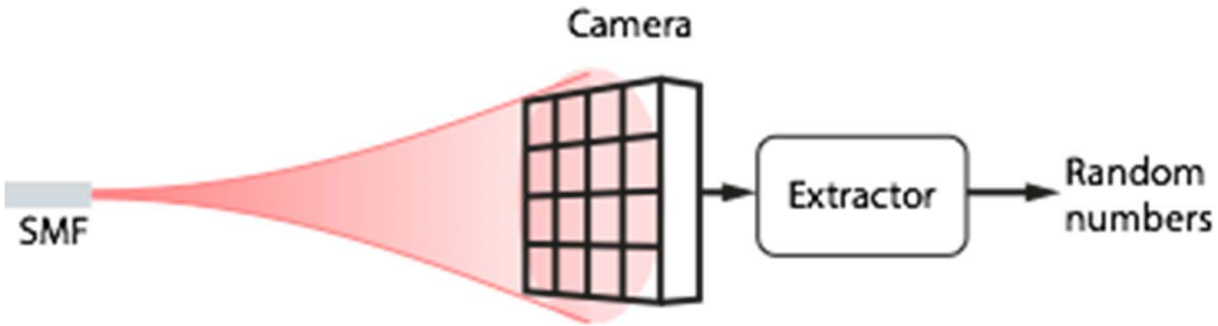


- Steganography exploiting shot noise



Example with a Nokia N10

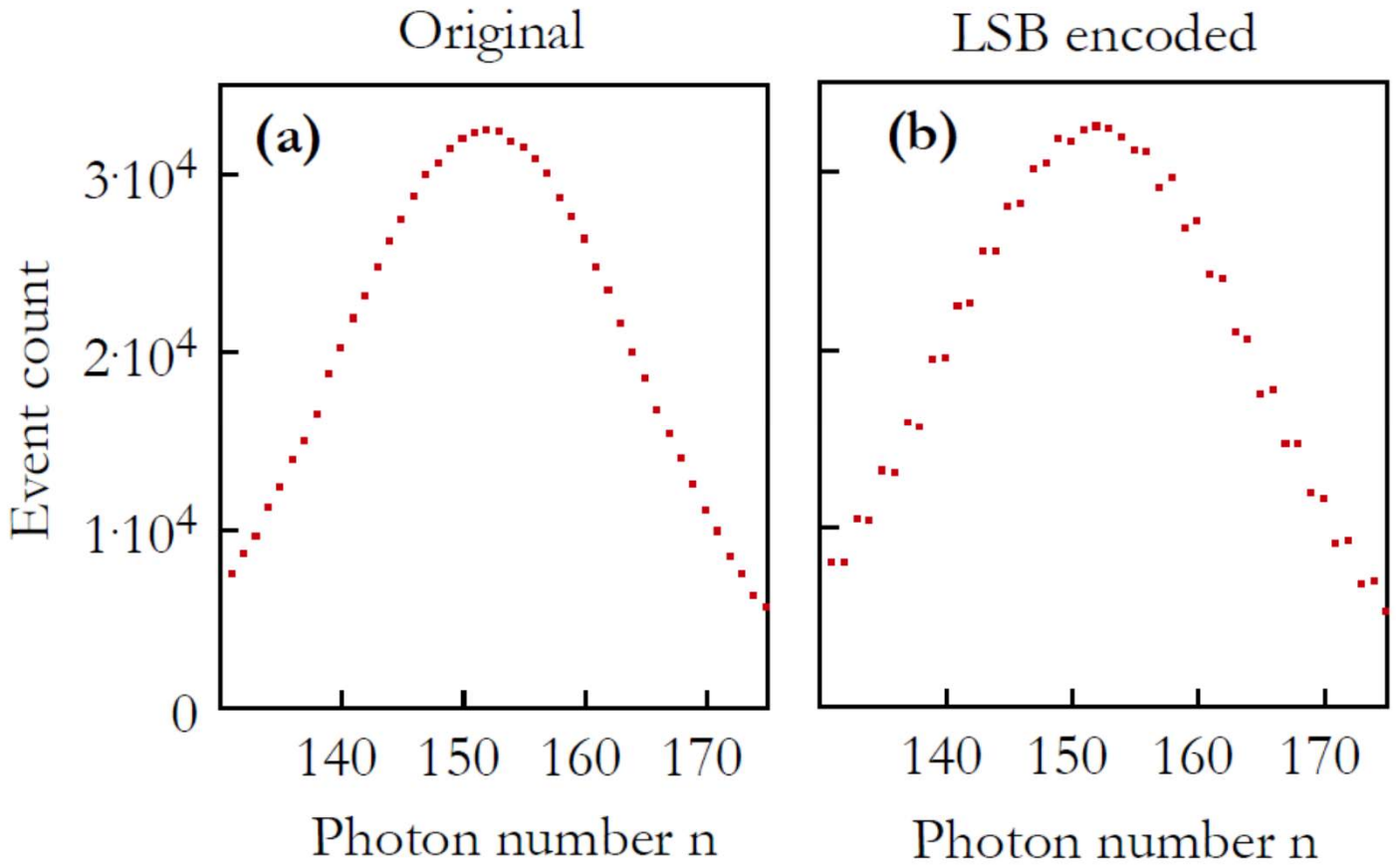




Sanguinetti B., *et al.* 2004 Phys. Rev. X 4 031056

Naive idea

- Use least significant bit to transmit (OTP) encoded data



Simulated Histogram of the pixel values of a homogeneous area

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Better idea

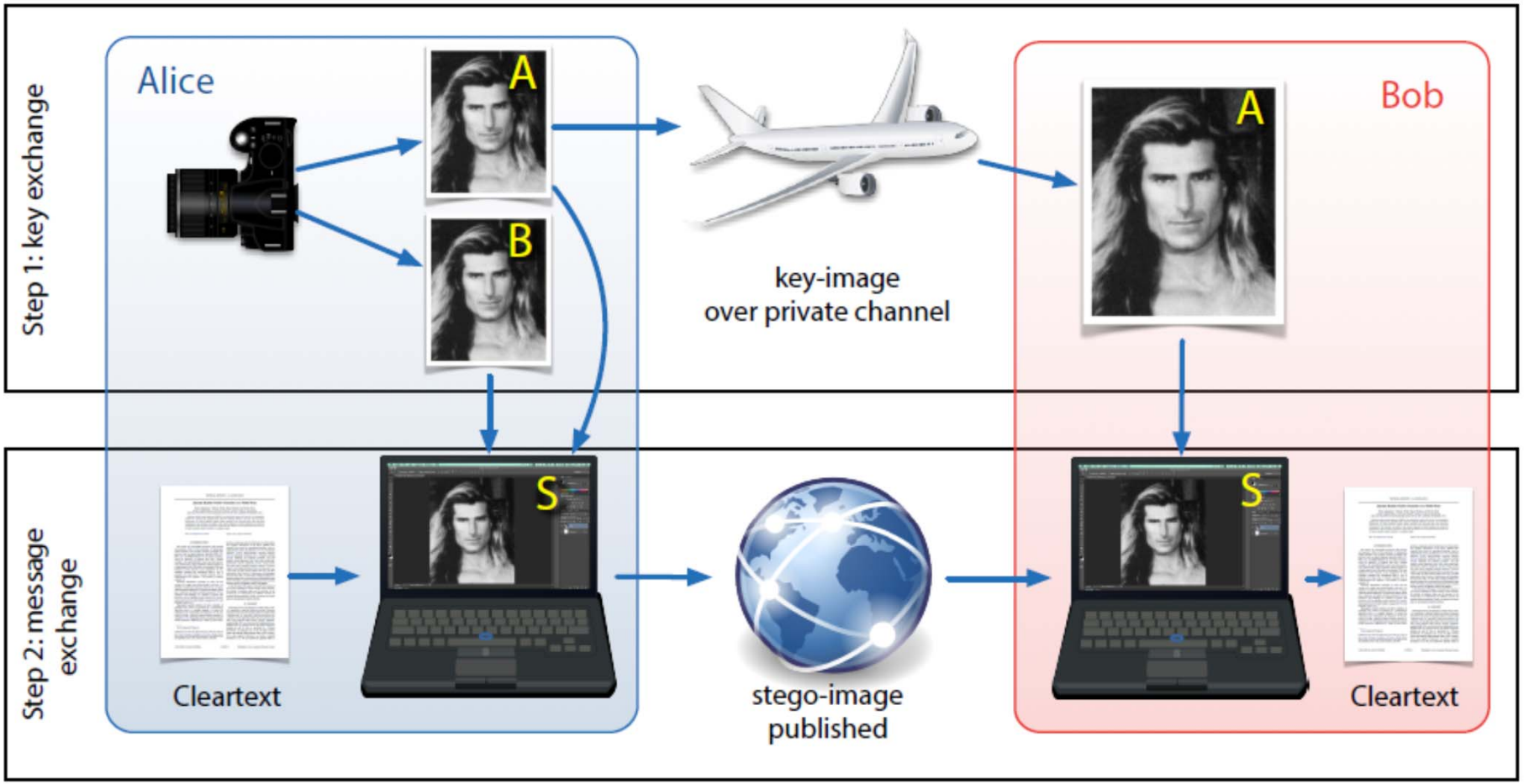
- Take photographs of a static object in rapid succession
- Assumptions:
  1. state of object and camera unchanged between to consecutive pictures K and C
  2. Each pixel is statistical independent (no crosstalk).

- Protocol: given Text T, create a new picture S as follows:

$$S_i := \begin{cases} K_i, & \text{if} \quad T_i = 0 \\ C_i, & \text{if} \quad T_i = 1 \end{cases} \qquad T_i := \begin{cases} 0, & \text{if} \quad S_i = K_i \\ 1, & \text{if} \quad S_i \neq K_i. \end{cases}$$

- S cannot be distinguished from any real photograph

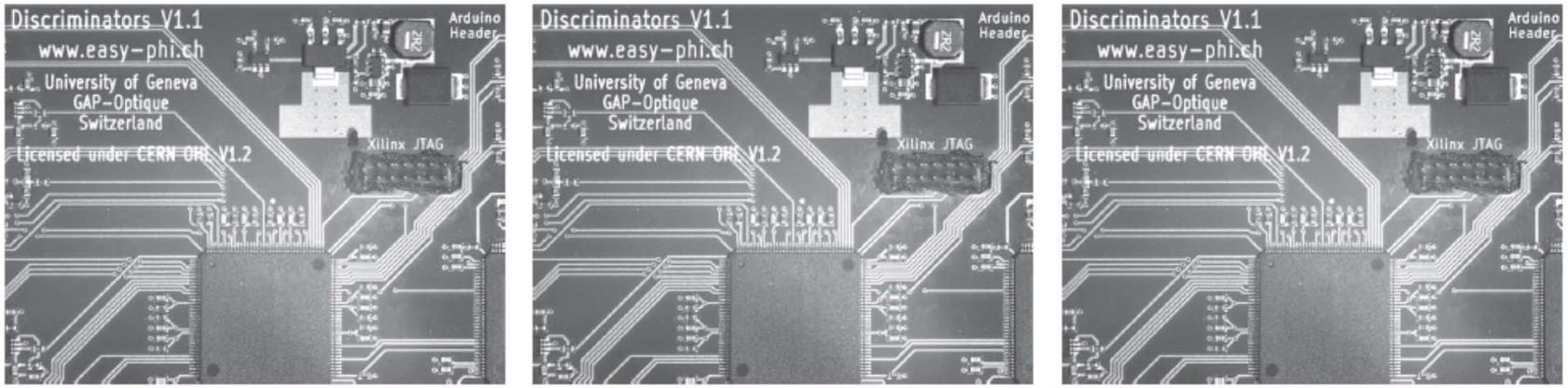
Private key steganography



Experimental realisation

- Tests with scientific mono-chrome and consumer colour cameras with raw image files
- 8 Mpix 16 bit tiff files

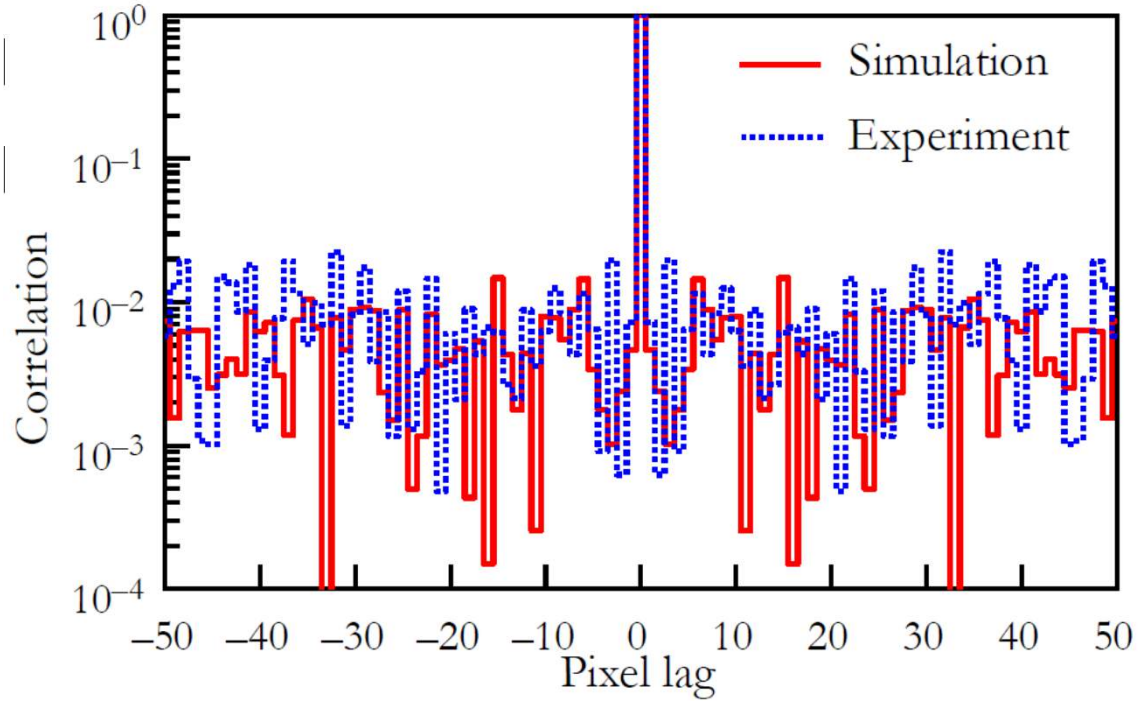
Key-image A | Auxiliary image B | Stego-image S



- error-correction applied (Reed-Solomon code)

Results

- It works!
  - no cross-pixel correlations
  - stability depends on experimental situation



stigations

bits can be

Conclusions



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Announcement: Winter School 2016

SWISS QUANTUM



ID Quantique is proud to announce its 8th Winter School

- Topic: Quantum Cyber Security;
- Date: Sunday, January 17th to Thursday, January 21st 2016;
- Location: Les Diablerets, Switzerland;
- Key speakers:

  John Martinis, Google; Michele Mosca, IQC, Canada;
  Gilles Brassard, Uni. Montréal; Nicolas Gisin, UniGe.



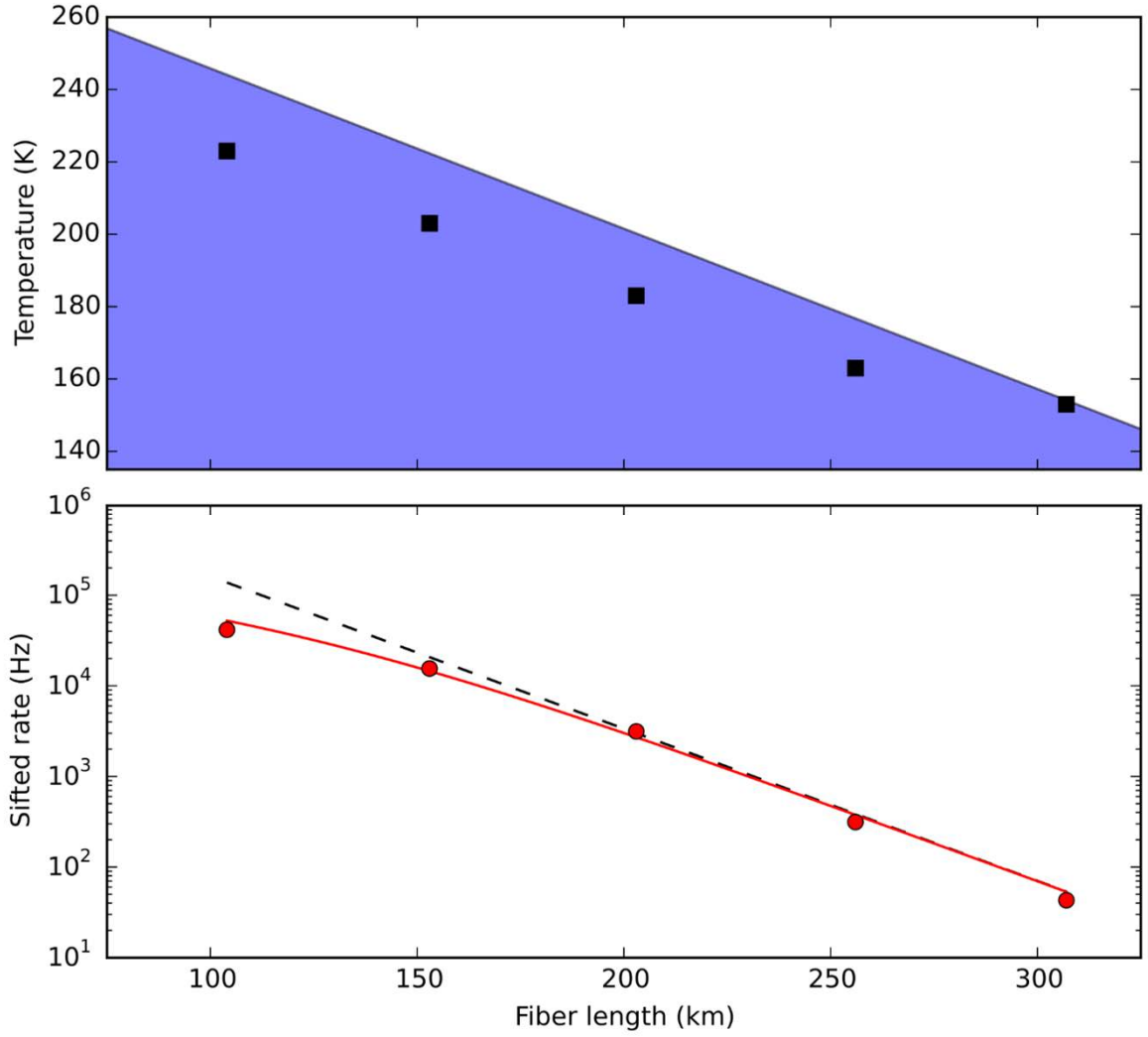
- More information on at: 8th-winter-school-on-quantum-cyber-security
- Contact: ws@idquantique.com

IDQ

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- Everyone has the right to freedom of opinion and expression; this right includes freedom to hold opinions without interference and to seek, receive and impart information and ideas through any media and regardless of frontiers.

Limiting temperature



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Stirling coolers





3.5 kg
153 K

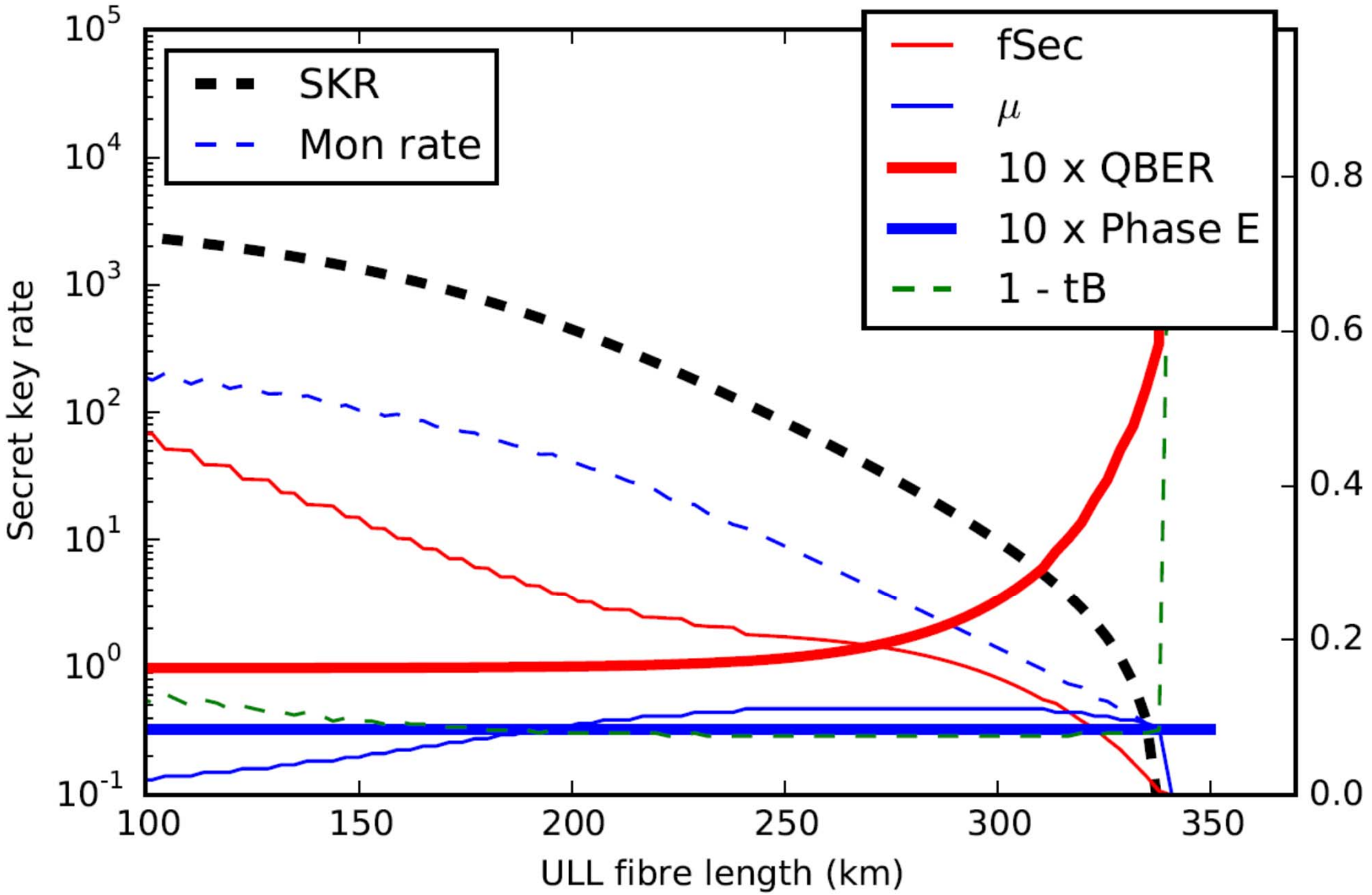
220 g
110 K

Jean-Yves Martin et al., "*Thales Cryogenics rotary cryocoolers for HOT applications*," Proc. of SPIE Vol. 8353 (2012)

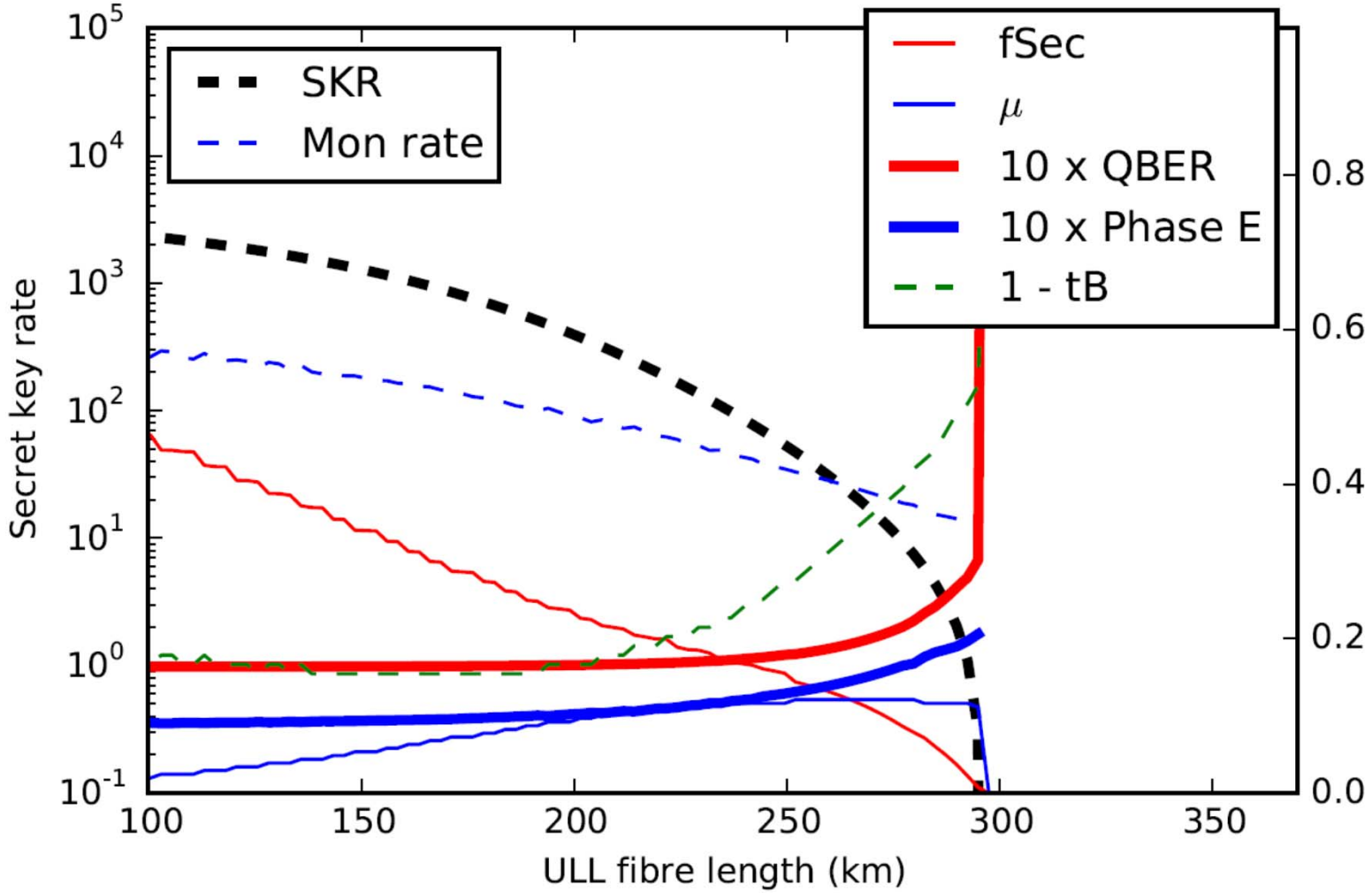
Optimize parameters for all distances!

Channel		Detectors							Distillation					Performance		
L (km)	α (dB)	T_{det} (K)	η_{dat} (%)	D_{dat} (cps)	τ_{dat} (μs)	η_{mon} (%)	D_{mon} (cps)	τ_{mon} (μs)	μ	n_{cpp}	t_{cpp}	f_{IR}	f_{sec}	$\hat{V}$ (%)	Q_{tot} (%)	r_{sec} (bps)
104	16.9	223	22	548	9	20	384	34	0.060	2.0×10^7	536.8	1.217	0.347	98.3	2.4	1.27×10^4
153	25.7	203	27	117	25	25	86.5	76	0.090	2.0×10^7	1450	1.272	0.310	98.0	1.5	5.20×10^3
203	34.1	183	27	16.2	42	25	14.4	89	0.100	1.1×10^7	3108	1.271	0.301	98.1	1.5	1.02×10^3
256	42.6	163	27	3.26	85	25	2.80	106	0.095	1.3×10^6	8586	1.228	0.230	98.2	2.0	78.0
307	51.9	153	22	1.33	114	20	0.87	114	0.075	6.6×10^5	17245	1.287	0.084	96.5	3.5	3.18

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